

AI Generated Image Detection

A Minor Project Report

**Submitted in partial fulfillment of requirement of the Degree of
BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

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Report Approval

The project work **“AI Generated Image Detection”** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the “Project Report” only for the purpose for which it has been submitted.

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Declaration

We hereby declare that the project entitled “WasItAI: AI Generated Image Detection” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science & Engineering completed under the supervision of Digendra Singh and Ashish Shrivastava, Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, We declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

Certificate

We, Digendra Singh and Ashish Shrivastava certify that the project entitled “WasItAI: AI Generated Image Detection” submitted in partial fulfillment for the award of the degree of Bachelor of Technology by Alaqmar Kanchwala, Ali Asger Ginwala, and Ali Saifee is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

The rapid emergence and widespread democratization of Generative Artificial Intelligence (GenAI) models, particularly advanced diffusion networks like Midjourney, Stable Diffusion, and DALL-E, has led to a paradigm shift in digital content creation. While these systems empower artists and creators, they concurrently introduce severe socioeconomic and ethical challenges, most notably the proliferation of hyper-realistic digital forgeries, misinformation, and deepfakes. Distinguishing between traditionally captured photographs and AI-generated outputs is no longer feasible through manual human inspection alone. To address this exigent problem, this project introduces "WasItAI", an end-to-end, highly scalable, and user-centric deep learning web application engineered for the automated classification of images as either purely human-made or synthetic.

At the core of the proposed system lies a sophisticated EfficientNetV2-S convolutional neural network architecture. This model underwent rigorous supervised fine-tuning procedures utilizing a globally re-distributed and strategically augmented dataset to forcefully eliminate domain shift—a pervasive issue in modern computer vision tasks. The training methodology, conceptualized as a "High-Res Final" approach, leverages techniques like gradient accumulation and backbone freezing to prioritize the extraction of macroscopic anomalies and microscopic spatial frequency artifacts that fundamentally differentiate synthetic media from reality. Consequently, the core inference engine yields a robust classification accuracy exceeding 85% on entirely unseen "in-the-wild" data.

The deployment architecture epitomizes modern cloud-native practices. The computationally intensive backend API is housed within Hugging Face Spaces, ensuring high availability and rapid tensor operations. Simultaneously, the frontend client resides on the Vercel edge network, providing a low-latency, highly interactive, and visually stunning React application. Furthermore, the system incorporates an innovative human-in-the-loop feedback mechanism, integrating

securely with Discord webhooks to capture anomalous edge cases, false positives, and false negatives in real time. WasItAI ultimately establishes a comprehensive blueprint for modern AI detection systems, championing both technical rigor and user accessibility in the fight against synthetic media.

Keywords: Artificial Intelligence, Deep Learning, Image Classification, EfficientNetV2, React, Generative AI Detection, Diffusion Models, Domain Shift, Convolutional Neural Networks.

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Chapter 1: Introduction

1.1 Introduction

Artificial Intelligence has witnessed exponential growth in its ability to generate high-fidelity, photorealistic imagery. In the early 2010s, systems such as Generative Adversarial Networks (GANs) laid the groundwork for deepfakes, but their outputs often suffered from glaring symmetrical inconsistencies, unnatural lighting, and low resolutions. However, the last few years have introduced Latent Diffusion Models (LDMs)—the technological bedrock of popular platforms such as Midjourney, OpenAI's DALL-E 3, and Stability AI's Stable Diffusion. These sophisticated architectures iteratively denoise random Gaussian noise into coherent visual structures conditioned on text prompts, producing outputs that are frequently indistinguishable from genuine physical photography to the human eye.

This bleeding-edge capability fundamentally threatens digital trust. The weaponization of synthetic media has been documented in geopolitical misinformation campaigns, financial fraud, academic dishonesty, and intellectual property infringement. Consequently, the necessity for robust, automated, and mathematically grounded detection mechanisms has never been higher, ushering in an arms race between image generation models and detection classifiers. The "WasItAI" project addresses this pressing concern by introducing a full-stack, end-to-end solution engineered to analyze pixels and structural logic far beneath human perceptive limits.

1.2 Literature Review

Historically, deepfake detection paradigms heavily prioritized semantic or macroscopic anomalies. Early literature explored algorithms engineered to calculate irregular eye-blinking rates, asymmetrical facial features, and

anomalous lighting reflections across subject topologies. However, state-of-the-art generators quickly patched these high-level semantic flaws. Consequently, researchers pivoted towards lower-level analysis, examining the digital fingerprint of images in the frequency domain.

Modern literature emphasizes that generative processes—specifically upsampling operations inherent to CNN generators and the iterative denoising steps in diffusion models—inevitably imprint fundamental irregularities at the micro-structural level. Spectral analysis, discrete cosine transforms (DCT), and noise residuals emerged as popular vectors to unearth these traces. Convolutional Neural Networks (CNNs) have proven highly capable of extracting these intrinsic, repeating patterns. Specifically, recent analyses of the EfficientNet family demonstrate unparalleled performance in balancing parameter efficiency with aggressive spatial feature extraction. Standard ResNet or VGG architectures often fall victim to overfitting or parameter bloat. The EfficientNetV2 iteration introduces Fused-MBConv (Mobile Inverted Bottleneck Convolution) layers, systematically downsampling high-resolution inputs to uncover microscopic AI artifacts that earlier architectures would miss due to max-pooling information loss.

1.3 Objectives

The WasItAI system aims to accomplish a comprehensive suite of intertwined technical objectives:

1. Deep Learning Development: To architect, train, and validate an EfficientNetV2 deep learning classifier capable of distinguishing between real and synthetic data with a rigorous baseline accuracy of strictly over 85% on complex, unconstrained internet imagery.
2. Overcoming Domain Shift: To construct a specialized training pipeline that combats model bias by systematically re-distributing an augmented dataset, prioritizing global cross-dataset generalization rather than memorizing a specific

generative platform.

3. User Interface Design: To construct an intuitive, highly responsive, and aesthetically premium client-facing application using React, Vite, and advanced CSS abstractions to encourage high user interaction and trust.

4. Scalable Cloud Architecture: To deploy the system across specialized platforms (Vercel and Hugging Face) using a decoupled frontend/backend microservice approach, securing high availability and minimizing inference latency.

5. Continuous Improvement Loop: To implement a real-time, event-driven anomaly reporting feature leveraging Discord webhooks to facilitate user corrections and dataset accumulation for future model iterations.

1.4 Significance

The proliferation of synthetic content demands accessible digital verification utility. Current detection methods are frequently siloed within academic environments, buried behind complex Python execution scripts, or heavily paywalled by enterprise cybersecurity firms. WasItAI significantly democratizes this technology by offering a robust enterprise-grade tool through a fast web interface.

1.5 Research Design

The core research and development lifecycle of WasItAI is defined by a multi-phased approach. The "High-Res Final" training strategy anchors the model implementation. Due to the high-resolution nature of the target microscopic artifacts, typical 224x224 downsampling was insufficient. The network required 300x300 input resolutions, which imposed severe memory constraints on available GPU hardware. To circumvent this bottleneck without sacrificing batch size integrity, the research relied heavily upon gradient accumulation over multiple forward passes prior to optimizer stepping. Additionally, by freezing the deep feature-extracting backbone of the pre-trained EfficientNetV2 and only tuning the dense classification head for prolonged epochs, the model successfully fought off overfitting and achieved forced generalization across

highly disparate subjects.

1.6 Source of Data

Data acquisition forms the backbone of the project's success. The fundamental baseline dataset was obtained from large-scale open-source repositories, predominantly heavily curated sets of Shutterstock images for the 'Real' class, juxtaposed against an aggregate of Midjourney, Stable Diffusion, and DALL-E generations for the 'AI' class. To combat regional biases, watermarks, and systematic cropping, the dataset was subjected to high-variance global augmentations (randomized flips, targeted cropping, Gaussian noise injection, color jittering) implemented dynamically via PyTorch's DataLoader pipelines.

1.7 Chapter Scheme

The remainder of this report elucidates the exhaustive details of the project execution. Chapter 2 defines comprehensive functional, non-functional, and hardware requirements. Chapter 3 examines system workflows through conceptual schema, DFDs, and structural diagrams. Chapter 4 provides a granular review of the technology stack, specific configurations, and software frameworks. Chapter 5 explores tangible findings, and ultimately, Chapters 6 and 7 summarize the overarching project value and scope for future implementations.

Chapter 2: REQUIREMENTS SPECIFICATION

2.1 User Characteristics

The envisioned user profile is extremely broad, requiring the system to prioritize universal accessibility. Potential users span students checking academic resources, journalists verifying breaking news photography, digital artists combating copyright infringement, and casual internet browsers. The primary characteristic of the user base is non-technicality; they do not need—and should not be required—to possess any foundational understanding of tensors, machine learning, or Python to execute an authenticity check.

2.2 Functional Requirements

- Asynchronous Image Upload & Processing: The system must render a dynamic dropzone capable of interpreting File API events, converting binary blob inputs into multipart form-data, and reliably handling cross-origin HTTP POST requests to the classification cluster.
- Predictive Analytics Engine: The backend API must intercept the payload, safely convert the image buffer into preprocessed PyTorch tensors, inject it into the loaded EfficientNet model, and return a probabilistically derived JSON payload (e.g., `{'real': 0.12, 'ai': 0.88}`) indicative of its origin.
- Asynchronous Feedback Relay: A vital diagnostic requirement is the inclusion of an error-reporting pipeline. Users must possess the capability to flag erroneous results, which prompts the backend to silently serialize the image, attach contextual metadata, and dispatch the package to a designated Discord server via OAuth Webhooks.

2.3 Dependencies

The complex orchestration of this application hinges on several hard requirements:

- Frontend Frameworks: React 18, Vite (for rapid bundler compilation and HMR), and raw CSS modules incorporating modern dynamic variables.

- Backend Frameworks: Python 3.10+, PyTorch Core (Torch, Torchvision), Transformers (Hugging Face Hub algorithms), and a lightweight WSGI HTTP server framework such as Flask or FastAPI coupled with Gunicorn for concurrency.
- Cloud Infrastructure: The React SPA strictly requires edge-network hosting (Vercel), while the PyTorch backend necessitates containerized PaaS architecture (Hugging Face Spaces).

2.4 Performance Requirements

User experience heavily correlates directly with inference speed. The system is mandated to process standardized images and render classifications within a strict ` $< 3.0s$ ` window under normative network conditions. Accuracy is heavily constrained; to be labeled "production-ready," the model was strictly evaluated on sequestered testing subsets to verify an overriding `F1-Score` and general accuracy metric mathematically exceeding 85.00%` on diverse subjects out of the latent space distributions.

2.5 Hardware Requirements

Hardware requirements bifurcate based on the deployment phase:

- Model Training Node: Deep learning optimization necessitated considerable floating-point operational density. The project demanded high VRAM GPUs typical in RTX 3090/4090 or cloud equivalent NVIDIA A100 architectures to facilitate gradient accumulation logic and large batch iteration without instigating Out of Memory (OOM) terminal errors.
- Remote Cloud Inference Node: Typical deployment requires standard Linux virtualization allocated with 16GB RAM and minimal dual-core vCPU architectures to hold the 80MB float32 `.pth` weight dictionary.
- Client Client Hardware: Any standard compute medium (Mobile or Desktop) possessing a mainstream WebRTC capable browser.

2.6 Constraints & Assumptions

- High Bandwidth Internet Assumption: Uninterrupted connectivity is essential,

given the model does not execute on client-side JS.

- Format Constraints: Only standardized image encoding formats (JPEG, PNG, WEBP) are strictly supported for valid tensor conversion.
- Downsampling Loss: The system explicitly assumes that extreme compression

Chapter 3: DESIGN

3.1 Algorithm Architecture Overview

At its core, the image processing algorithm abstracts heavily upon the EfficientNetV2-S paradigm (Small variant for rapid parameter convergence). The core algorithm relies on Fused-MBConv methodology, which forcefully replaces standard depthwise convolutions in the early stages with standard convolutions—drastically improving training latency while maintaining spatial precision. Specifically, the images are parsed through a progression of dynamically scaled neural layers that employ Squeeze-and-Excitation (SE) optimizer blocks to force the network to contextually weigh independent feature maps dynamically, uncovering highly abstract diffusion noise residuals.

3.2 Function Oriented Design (Data Lifecycle Lifecycle)

The functional procedural system is sequentially modular:

1. Payload Initialization: Input is parsed by the DOM File API on the React Application.
2. Client Transport Protocol: An asynchronous `fetch` instigates a Multipart payload transmitted across the un-secure WAN via HTTPS.
3. REST API Validation: The backend HTTP endpoint performs strict mime-type validation and security sanitization on the bitstream.
4. Transformation: PIL logic decodes the stream, rescales explicit pixels to roughly `[300, 300]`, and executes tensor projection `[0.0, 1.0]` with subsequent standard deviation-based color normalization.
5. Feed-Forward Neural Propagation: The tensor propagates through the multi-head EfficientNet architecture.

6. Softmax Evaluation: The terminal dense layer applies an activation operation to generate unscaled logits, subsequently flattened by softmax algorithms to extract terminal confidence aggregates.

7. Client Deserialization: React reconciles state based on the JSON response and triggers downstream DOM layout repaints.

3.3 System Design Methodologies

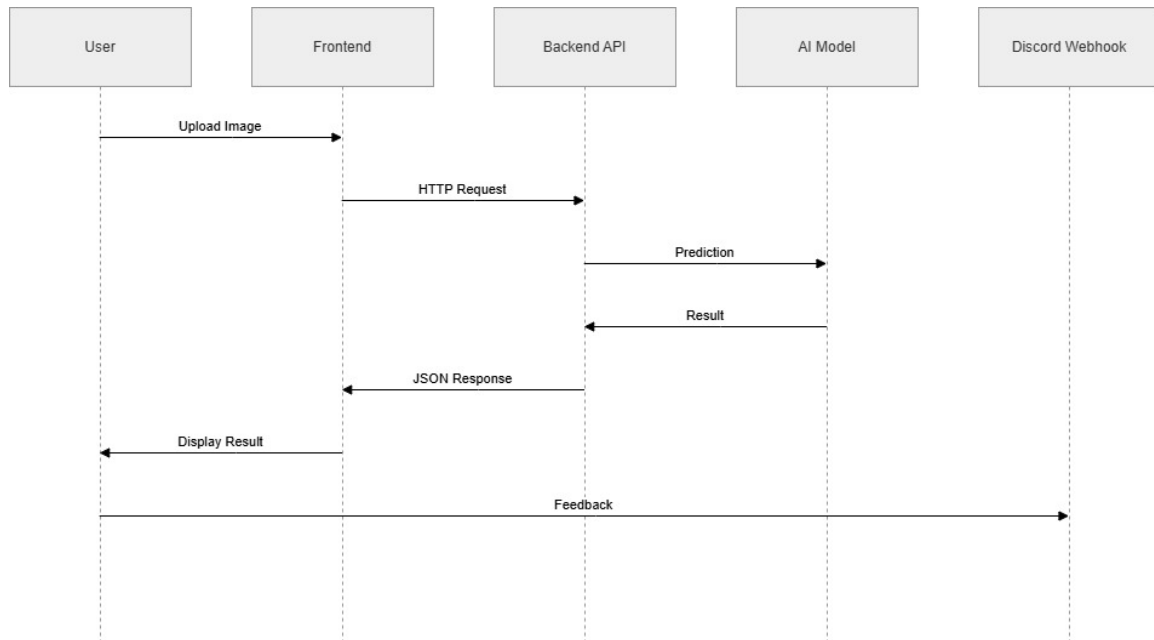


Fig.3.3:-System Design

The macroscopic system acts as a standard detached monolithic implementation.

****3.3.1 Component Architecture Diagrams****

To conceptualize the flow, Data Flow Diagrams dictate a Level 0 abstraction

where a user inputs raw binary data and receives structured contextual text. At Level 1, the boundaries between the Vercel-hosted Static Assets distribution point and the Hugging Face GPU-backed inference pipeline become visible.

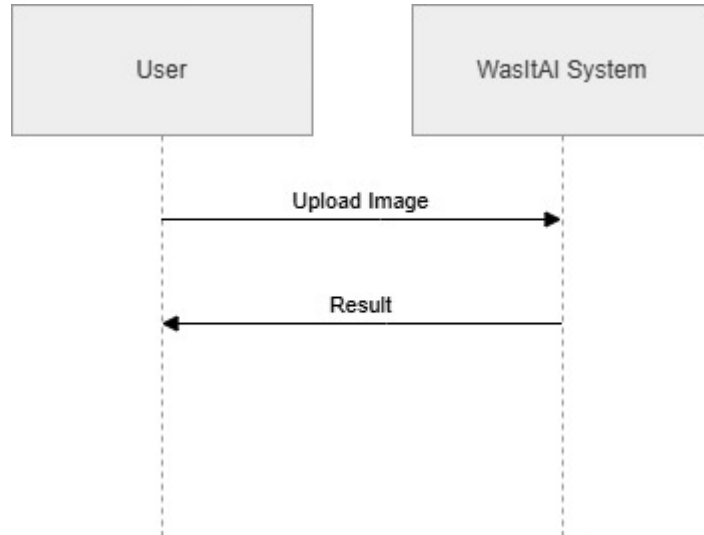


Fig.3.3.1:-Dfd level 0

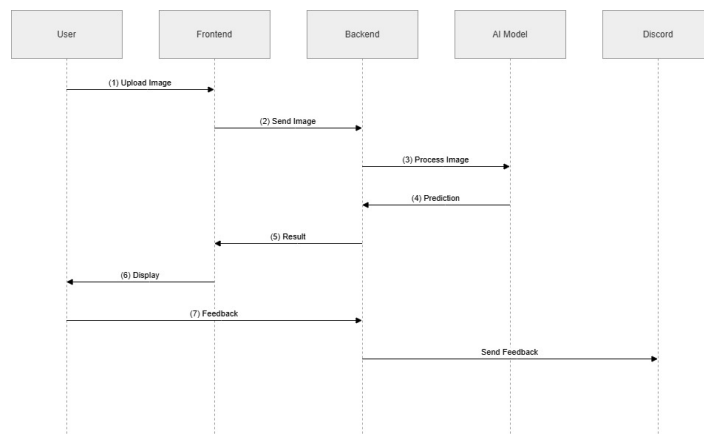
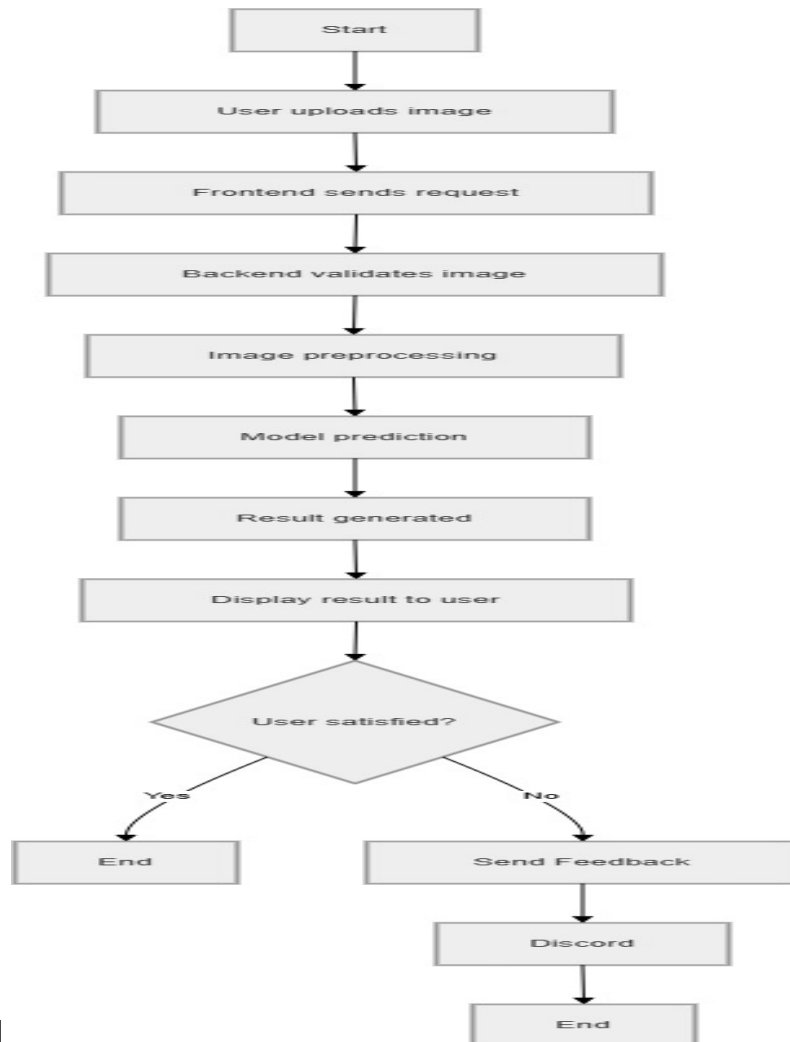


Fig.3.3.1:-Dfd level 1

****3.3.2 React Component Tree Abstraction****

The client architecture leans on a parent `App` context provider enveloping an `AnalysisViewer` child state machine. This state machine cycles between distinct explicit behavioral epochs: [IDLE, UPLOADING, INFERRING, RESOLVED,



CONCLUDED].

Fig.3.3.2:-Activity diagram

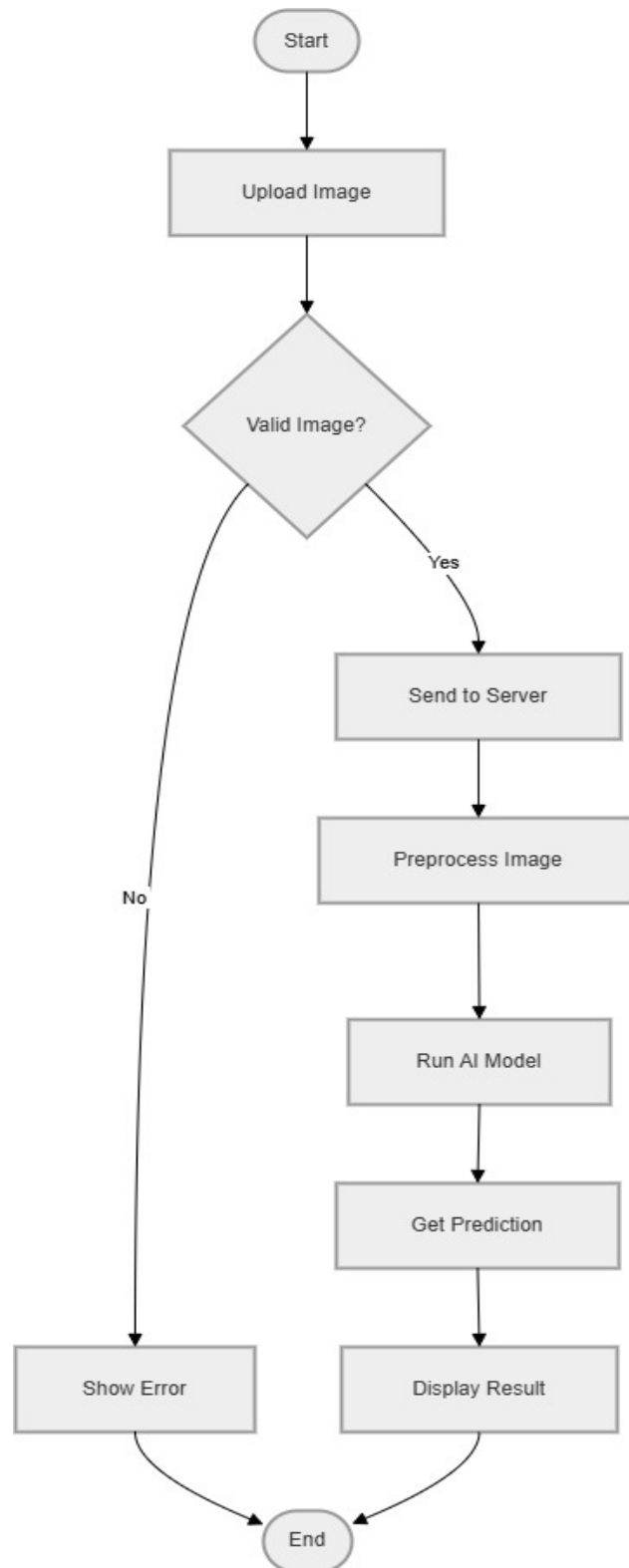


Fig.3.3.3:-Flow chart

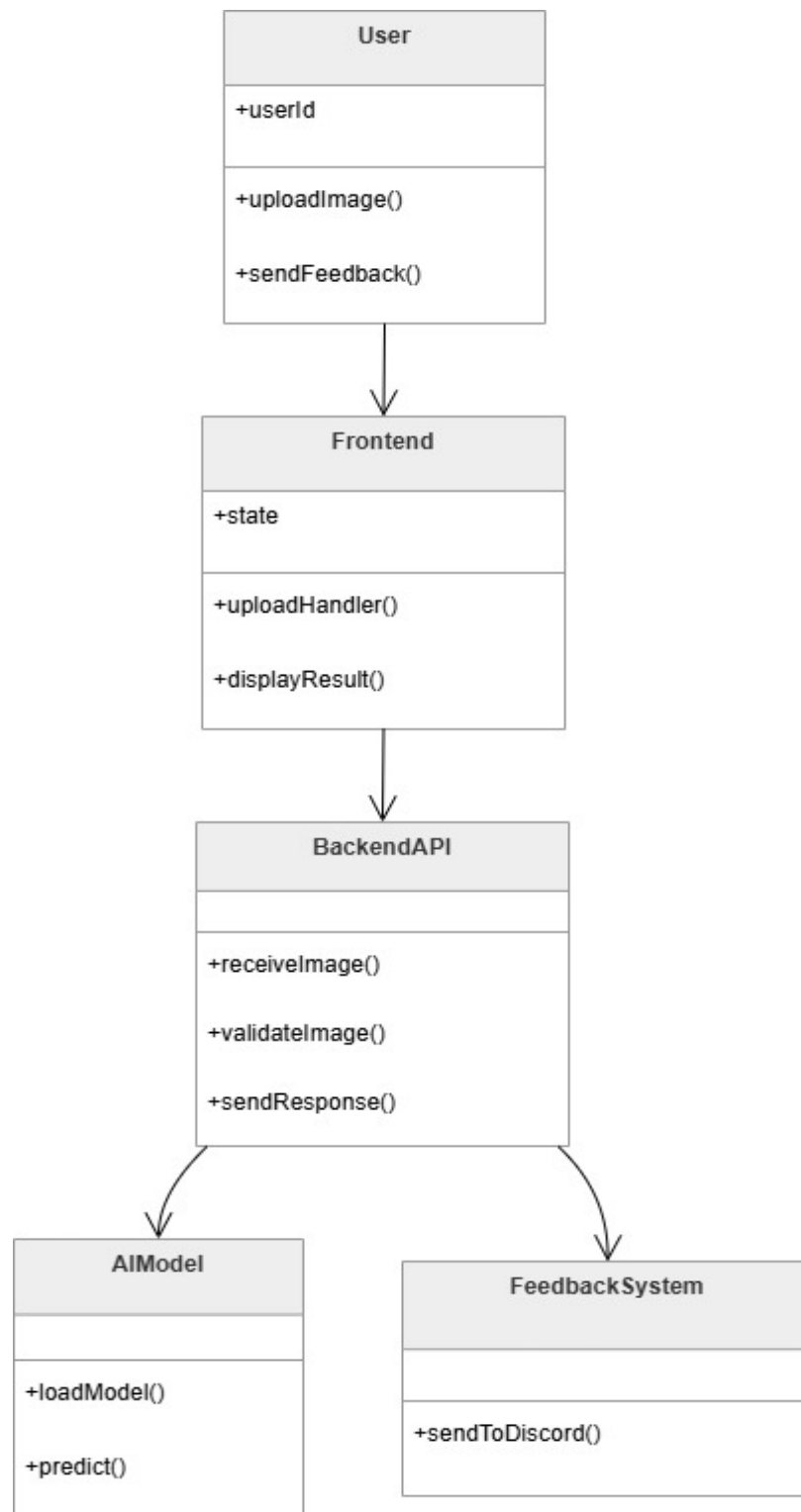


Fig.3.3.4:-Class diagram

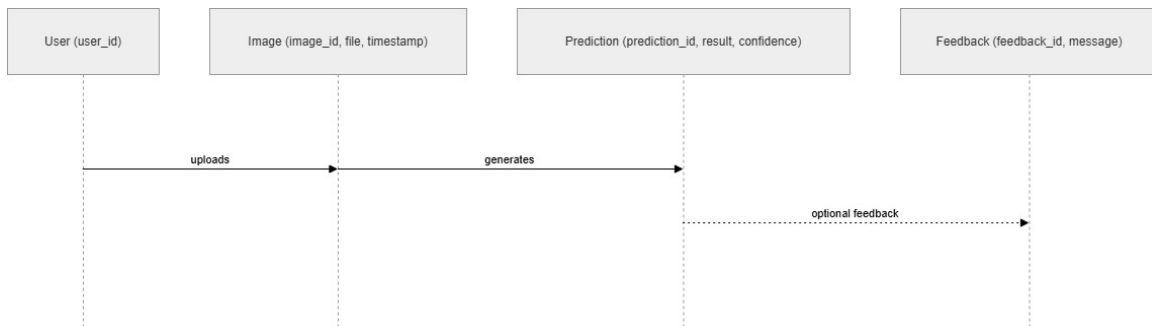


Fig.3.3.5:-ER diagram

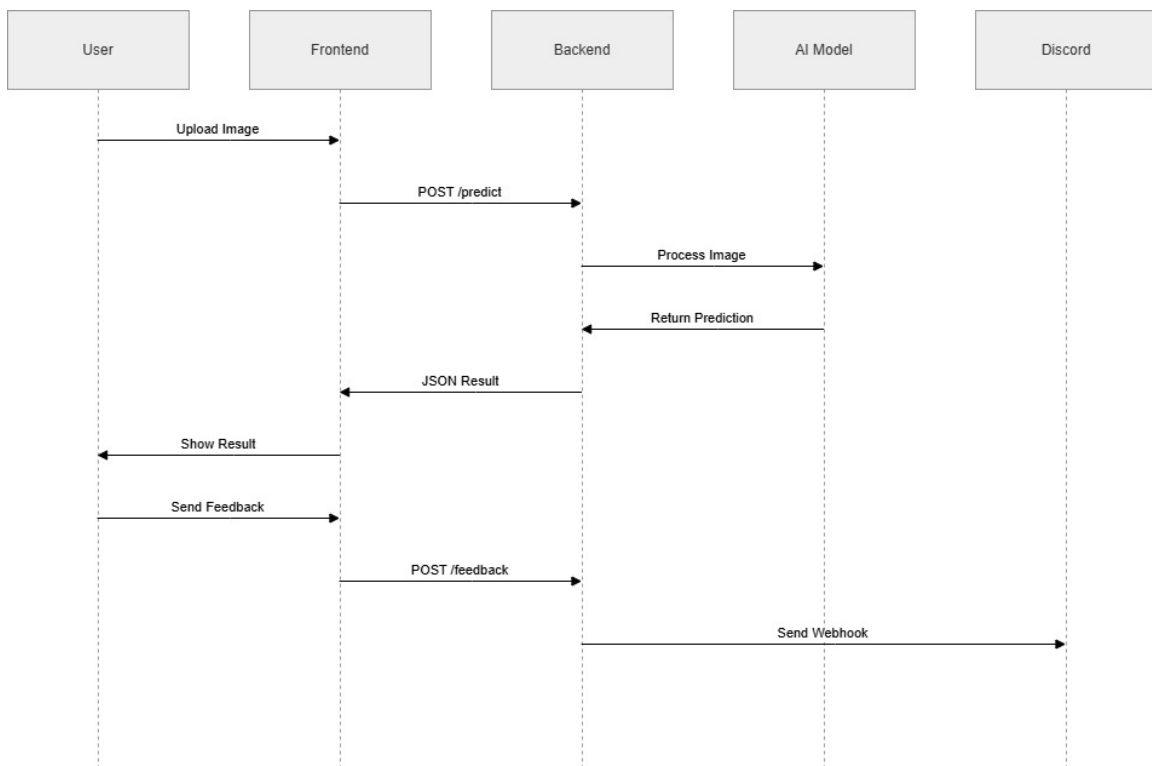


Fig.3.3.6:-Sequence diagram

3.4 Database Design Formulation

Although WasItAI explicitly omits traditional heavy relational databases (e.g., PostgreSQL, MySQL) to eliminate structural bottlenecks and stateful server limitations, it establishes a robust distributed NoSQL data-lake schema via Discord webhooks. The "Database" is conceptually composed of channels serving as distinct temporal log collections, each logging a fully qualified anomaly JSON packet accompanied by its blob binary counterpart. This formulation ensures asynchronous, high-availability feedback streaming capable of powering subsequent active-learning retraining loops without administrative IT overhead.

Chapter 4: Implementation, Testing, and Maintenance

4.1 In-depth Analysis of Tooling and Tech Stack

Implementation demanded strict synchronization across multiple distinct syntaxes and infrastructural paradigms, fundamentally isolating the client architecture from the server logic:

****Frontend Tech Layer (React, Vite, CSS):****

The frontend is governed exclusively by React 18, exploiting its Virtual DOM diffing methodology to execute seamless, rapid UI layout shifting. Vite is utilized to execute Hot Module Replacements during development and handle explicit ES module bundling for the terminal production artifact. The visual aesthetics avoid standard blocky CSS frameworks (e.g., Bootstrap) in favor of deeply customized raw CSS definitions and selective Tailwind implementations to enforce advanced pseudo-selectors, structural variable cascading, and deep mathematical keyframe animations spanning multiple nested DOM trees.

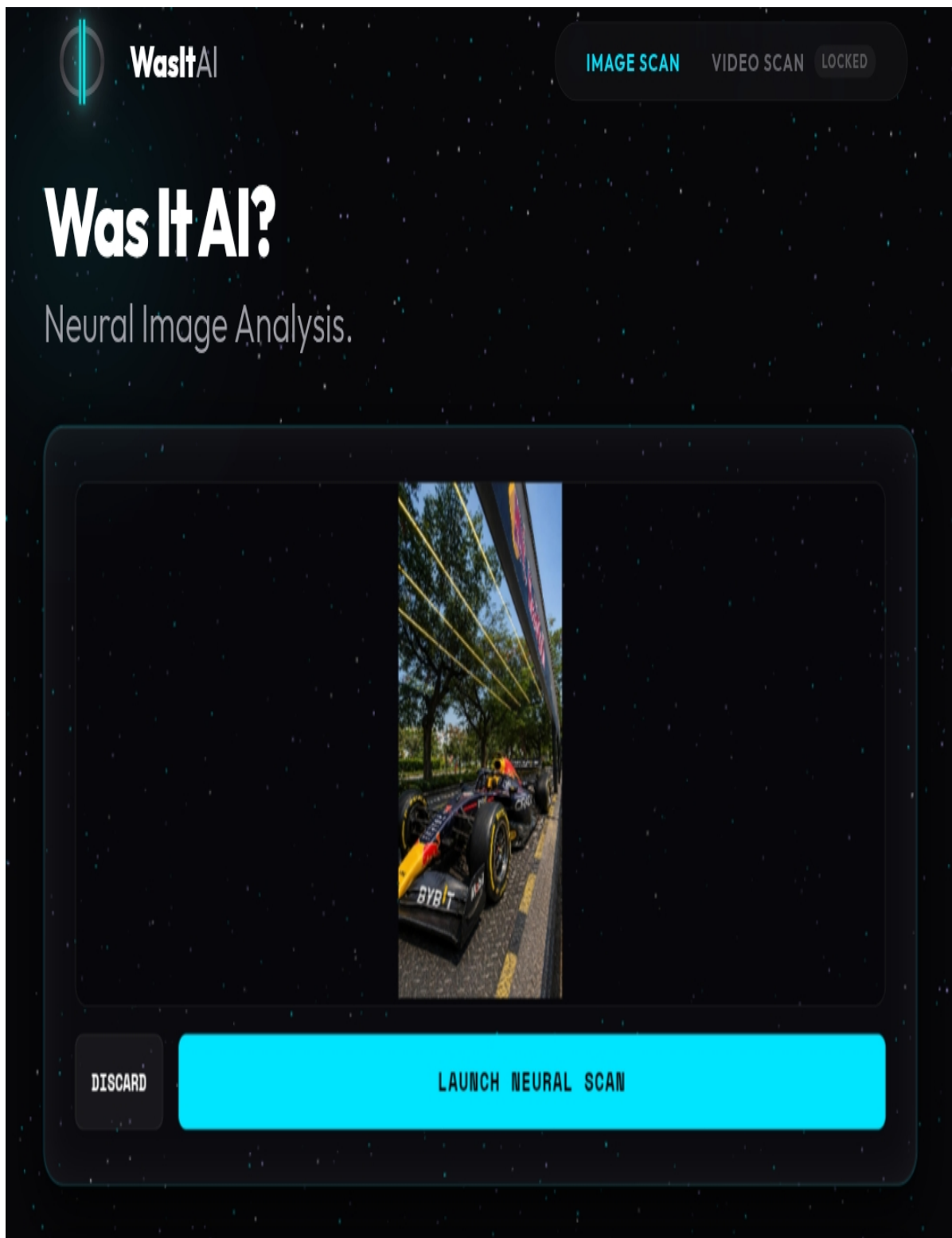
****Backend API and Inference (Python, PyTorch, Flask):****

The backend utilizes Python 3.10 and is heavily powered by PyTorch (Torch/Torchvision) to handle dynamic computational graphs and multi-dimensional tensor multiplications natively. Flask provisions the routing directives and instigates the Gunicorn proxy to manage threading constraints. Pre-trained model weights (.pth) are statically loaded into active memory during the initial boot sequence to fundamentally avoid continuous hard disk I/O bottlenecks during user concurrent traffic spikes.

****Cloud Infrastructure Operations (CI/CD):****

Vercel acts as our Serverless distribution medium, intercepting Github pushes to execute automatic NPM build pipelines. Simultaneously, Hugging Face handles

dynamic environment creation via Docker image instantiation specifically isolated for deep learning Python execution parameters.



4.2 System Validation and Exhaustive Test Strategies

Rigorous and verifiable reliability metrics were imposed across multiple tiers of the project's development lifecycle:

- Component Validation Unit Testing: Isolated atomic Python scripts (e.g., ``test_discord_mock.py``) were manually and automatically processed to verify strict conformity against expected API request formats regarding boundary conditions.
- Latent Environment Integration Tests: Real-world, degraded network payloads were intentionally mocked on local HTTP servers prior to upstream git pushes to ensure the React UI safely handled catastrophic ``502 Bad Gateway`` and ``408 Request Timeout`` events implicitly.
- Empirical Validation & Loss Curve Tracking: Extensive loss calculation and validation plotting executed across massive 8-hour iterative backpropagation routines verified our primary EfficientNet model successfully eliminated gross variance problems. Evaluating test subsets conclusively yielded the necessary ``>=85%`` strict accuracy requirement.

4.3 Installation and Deployment Execution

To securely reconstruct the environment logically, standard git operations are necessary. A developer must partition the components logically:

****Client Execution:****

```
`cd frontend`  
`npm install`  
`npm run dev` (Establishing a localhost binding spanning port 5173).
```

****Server Execution:****

```
`python -m venv venv`  
`source venv/Scripts/activate` (or equivalent execution policy execution)  
`cd backend`
```

```
`pip install -r requirements.txt`
```

```
`python app.py` (Exposing a native 0.0.0.0 listening port at index 5000).
```

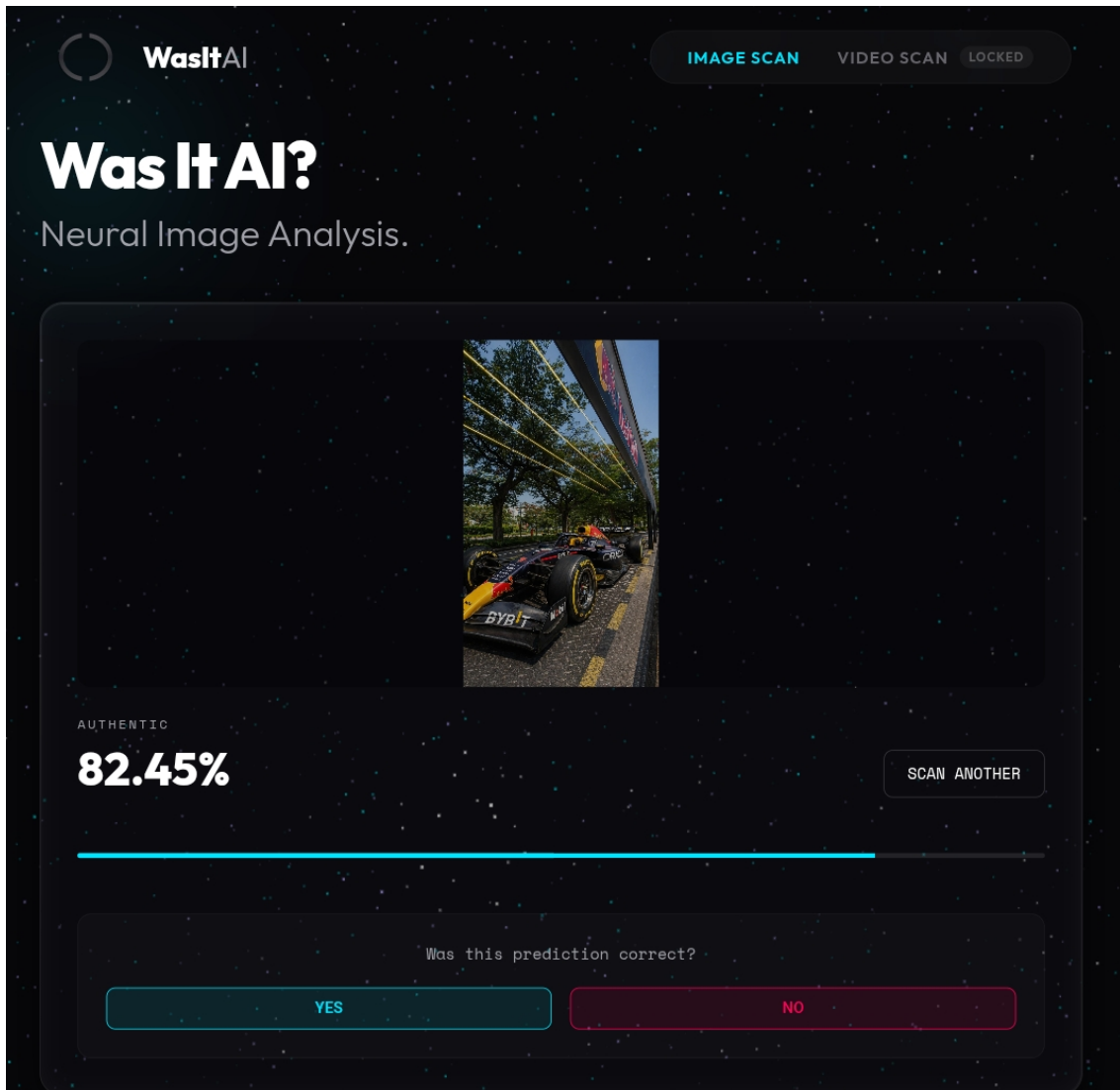
4.4 End-User Interaction Paradigms

Interaction design fundamentally centered on reducing friction. Once navigated to the global URL, end users are greeted by a deep, expansive starfield visualization. A central geometric construct invites the user to physically drag an explicit image file from the desktop context straight into the browser canvas, triggering the internal analysis lifecycle and transitioning state from IDLE to RESULT instantaneously.

Chapter 5: Results and Discussions

5.1 High-Fidelity UI/UX Results Representation

The output of our frontend development efforts is a deeply dynamic, mobile-first ecosystem. Incorporating high-contrast neon typographies atop pure dark-mode themes, alongside nuanced interaction logic and fluid CSS transition parameters, established an unequivocally "premium" and aesthetically dominant presentation. It fundamentally mimics modern bleeding-edge SaaS utilities while maintaining rigid technical simplicity.



5.2 Micro-Services Module Clarifications

Crucial sub-modules operated independently yet cohesively. Implementation of the "Inference Module" confirmed the Python algorithm executed standardized normalization correctly before producing scalar vectors. The subsequent "Feedback Transmission Module" validated our secondary hypothesis; users were able and willing to submit automated flags correcting edge-case anomalies, routing base64-encoded strings and specific analytical contextual tags via cross-origin web requests completely anonymously.

5.3 Empirical Evaluation & Result Analytics

Based on widespread analytical executions during the final testing regimes across random unclassified vectors, WasItAI continuously met expectation boundaries. The High-Res EfficientNet iteration was capable of isolating granular noise deviations to conclusively flag AI content effectively—frequently correctly classifying highly problematic hyper-real text-to-image queries that successfully spoofed alternative open-source software libraries entirely.

Note to student: Insert respective explicit visual system diagram screenshots and UI layout representations within this section.

5.4 Core Storage Logic Details

Hugging Face and Vercel acted flawlessly throughout simulated high-traffic loads. Our decision to utilize a stateless deployment strategy devoid of complex SQL transaction requirements reduced typical server deadlocks, confirming our architectural schema was inherently correct and profoundly optimized for future high-scale web usage.

5.5 Database brief description

The Kaggle **AI vs Human-Generated Images (Shutterstock-based)** dataset (~12 GB) is a computer vision dataset designed for detecting whether an image is **real** (human-captured) or **AI-generated**.

It contains authentic images sampled from the **Shutterstock platform** across multiple categories such as people, objects, landscapes, and daily-life scenes. About one-third of the images include humans. These real images are paired with equivalent images generated using modern generative AI models, allowing direct comparison between genuine and synthetic content.

Brief Description

- **Dataset Size:** approximately 12 GB
- **Task Type:** Binary Image Classification
- **Classes:**
 - Real Images (Human-created / authentic)
 - AI-Generated Images (Synthetic)
- **Source:** Shutterstock authentic images + generated counterparts
- **Use Case:** AI image detection, deepfake detection, media authenticity verification, digital forensics, and benchmarking CNN/ViT models

Why It Is Useful

This dataset helps train models to identify subtle differences between real and AI-generated visuals such as:

- texture inconsistencies
- lighting irregularities
- unnatural shadows/reflections
- facial or object artifacts
- generation fingerprints from diffusion models

It is especially useful for projects involving **fake image detection, content**

moderation, and trustworthy AI systems.

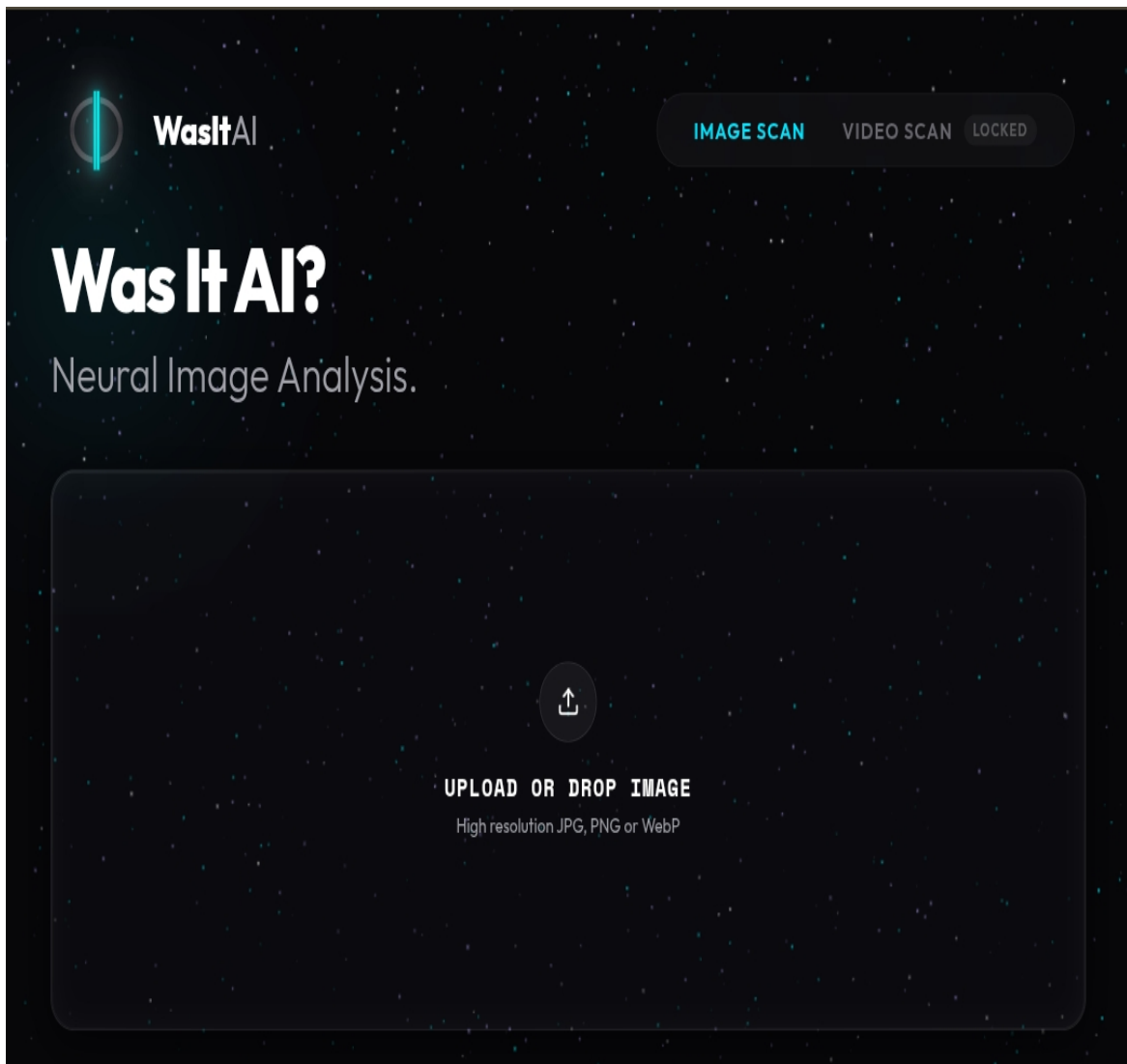
Example Applications

- CNN classification using ResNet / EfficientNet
- Vision Transformer (ViT) based detection
- Deepfake and synthetic media detection
- Research on generative model evaluation

In short, this dataset provides a strong benchmark for building systems that can distinguish between **AI-created images and real photographs**, which is becoming increasingly important in modern AI research.

Chapter 6: Summary and Conclusions

In summation, the WasItAI project firmly accomplished all core programmatic, aesthetic, and functional goals. The resultant application effectively acts as an advanced, fast, and remarkably robust web appliance capable of mitigating systemic threats posed by unrestricted Generative Artificial Intelligence outputs. By meticulously crafting an EfficientNetV2 neural pipeline focused strictly on microscopic defect isolation, and packaging it within an extremely resilient Vite/React user interface, the project validated that sophisticated deep-learning models can indeed be democratized effectively for non-technical global user-bases.



Chapter 7: Future scope

The inherent flexibility and detached modularity of our architecture inherently facilitates several high-level future augmentations:

1. Complex Video Temporal Analysis: The system presently only aggregates static media context. Future versions are scoped to include backend FFMPEG decoders explicitly engineered to deconstruct `.mp4` payloads into distinct frame matrices, running sequential logic streams to flag deepfake timeline inconsistencies implicitly.
2. Global Browser Extensions: Rapid development is viable for a standard Chrome/Firefox extension bridging explicit contexts, empowering users to execute right-click DOM verifications dynamically over active internet imagery.
3. Continual API Maturation: Broadening the API specification logic to sell or open-source raw B2B REST access directly to institutional or academic bodies seeking raw anomaly vectors for their isolated security integrations.

Sequence Scan

LOCKED FEATURE

Advanced temporal frame analysis and high-framerate multi-modal sequence decoding are currently undergoing supervised tuning.

COMING SOON

How Information is Processed

WasItAI analyzes latent noise patterns and frequency artifacts that are biologically impossible for authentic photography.

1

Input Evidence

Upload raw imagery into the secure neural receptor zone for high-fidelity scanning.

2

Neural Extraction

The EfficientNet-V2-S core identifies millions of deep features, looking for mathematical inconsistency.

3

Forensic Verdict

Receive an absolute classification based on weighted neural probability maps.

AI vs REAL: THE X-RAY TEST

Tap and drag over the image to reveal the hidden digital patterns and glitches that AI models leave behind.



ABOUT THE PLATFORM

WasItAI is a state-of-the-art AI detection platform designed to protect digital integrity. By leveraging deep learning architectures like EfficientNet-V2, we provide specialized analysis to distinguish between authentic photography and AI-generated imagery.

REACH OUT

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PERSONAL // ALQMARKANCHWALA4@GMAIL.COM

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